

LearnHub

Document 3: Requirements Gathering

Use Cases / Traditional Requirements / User Stories

Prepared by: Eman Al-Aodat | Mobile Application Development Course

1. Traditional Requirements

Traditional requirements document the complete set of functional and non-functional needs of the LearnHub application in a structured, specification-style format.

1.1 Functional Requirements

ID	Feature	Description
FR-01	Splash Screen	The app shall display a branded splash screen for 2 seconds on every launch.
FR-02	Onboarding Flow	A 3-step onboarding dialog shall be shown only on the first launch, stored via SharedPreferences.
FR-03	Course Listing	The home screen shall display a list of 8 available courses with title, category chip, rating, and instructor.
FR-04	Search	A real-time search bar shall filter courses by title as the user types.
FR-05	Category Filter	Filter chips (All, Programming, Design, Data Science, Business) shall filter the course list by category.
FR-06	Course Detail	Tapping a course shall open a detail screen showing description, lessons count, rating, and progress.
FR-07	Enrollment	Users shall be able to enroll in or unenroll from any course via a button on the detail screen.
FR-08	Progress Tracking	An interactive slider (0–100%) on the detail screen shall allow users to set their completion percentage per course.
FR-09	My Courses	A dedicated screen shall display only enrolled courses with progress bars and summary statistics.
FR-10	Profile Editing	Users shall be able to edit their display name with form validation (min 2 characters, not empty).
FR-11	Dark Mode	A toggle in Settings shall switch between dark and light themes app-wide, persisted across sessions.
FR-12	Data Persistence	All state (enrollments, progress, username, theme) shall be saved to SharedPreferences and reloaded on app start.
FR-13	Clear Data	A 'Clear All Data' option shall reset all stored preferences to their defaults after user confirmation.

1.2 Non-Functional Requirements

ID	Category	Description
NFR-	Performance	The app shall launch and display the home screen within 2 seconds

01		on a standard mobile device.
NFR-02	Usability	Navigation shall require no more than 2 taps to reach any primary screen.
NFR-03	Reliability	All data shall persist reliably across app restarts without data loss or corruption.
NFR-04	Portability	The app shall run on Android (API 21+) and iOS (13.0+) from a single Flutter codebase.
NFR-05	Maintainability	Code shall follow Flutter best practices using Provider for state management.
NFR-06	Offline Availability	All core features shall function without an internet connection.

2. Use Cases

Use cases describe the interaction between the primary actor (Student/User) and the LearnHub application to accomplish specific goals.

UC-01: Browse & Search Courses

Use Case ID	UC-01
Name	Browse & Search Courses
Actor	Student (User)
Precondition	App is launched and home screen is displayed.
Main Flow	1. User opens the app. 2. Home screen loads with all 8 courses. 3. User types in the search bar – course list filters in real time. 4. User taps a category chip – list filters by that category. 5. User taps a course card to open the detail screen.
Alternative Flow	If no courses match the search, a 'No results found' message is displayed.
Postcondition	User has found and opened a course of interest.

UC-02: Enroll in a Course

Use Case ID	UC-02
Name	Enroll in a Course
Actor	Student (User)
Precondition	User has opened a course detail screen.
Main Flow	1. User views course details (description, lessons, rating). 2. User taps the 'Enroll' button. 3. System marks the course as enrolled and saves to SharedPreferences. 4. Enrollment badge appears on the course card in the home screen.
Alternative Flow	If already enrolled, button displays 'Unenroll'. Tapping removes the course from enrollments.
Postcondition	Course enrollment status is saved locally and reflected across all screens.

UC-03: Track Learning Progress

Use Case ID	UC-03
Name	Track Learning Progress
Actor	Student (User)

Precondition	User is enrolled in at least one course.
Main Flow	1. User navigates to a course detail screen. 2. User drags the progress slider to a value between 0–100%. 3. System saves the progress value to SharedPreferences. 4. My Courses screen updates the progress bar and summary stats.
Alternative Flow	If user sets progress to 100%, the course is marked as completed in My Courses.
Postcondition	Progress is persisted and visible on My Courses screen.

UC-04: Customize Profile & Settings

Use Case ID	UC-04
Name	Customize Profile & Settings
Actor	Student (User)
Precondition	User is on the Profile screen.
Main Flow	1. User taps the edit icon next to their name. 2. User types a new name in the text field. 3. System validates the name (not empty, min 2 chars). 4. User saves – name is updated and persisted. 5. User toggles Dark Mode switch – theme changes app-wide instantly.
Alternative Flow	If name is invalid, an error message is shown and saving is blocked.
Postcondition	Username and theme preference are saved and applied.

3. User Stories (Jira/Trello Format)

User stories follow the standard Agile format: As a [role], I want to [action], so that [benefit]. These are organized as backlog items suitable for Jira or Trello.

ID	User Story	Priority	Status
US-01	As a new user, I want to see an onboarding guide on first launch, so that I understand the app features.	High	Done
US-02	As a student, I want to browse all available courses on the home screen, so that I can find topics I want to learn.	High	Done
US-03	As a student, I want to search for courses by name, so that I can quickly find a specific course.	High	Done
US-04	As a student, I want to filter courses by category, so that I can explore courses in a specific subject area.	Medium	Done
US-05	As a student, I want to view full course details before enrolling, so that I can make an informed decision.	High	Done
US-06	As a student, I want to enroll in a course with one tap, so that I can add it to my learning list.	High	Done
US-07	As a student, I want to unenroll from a course I no longer want, so that my course list stays relevant.	Medium	Done
US-08	As a student, I want to track my progress on each course using a slider, so that I know how much I have completed.	High	Done
US-09	As a student, I want to see all my enrolled courses in one place with progress bars, so that I can monitor my overall learning.	High	Done
US-10	As a user, I want to set my display name, so that the app feels personalized to me.	Medium	Done
US-11	As a user, I want to toggle dark mode, so that I can use the app comfortably in low-light environments.	Medium	Done
US-12	As a user, I want all my data to be saved locally, so that my progress is preserved after closing the app.	High	Done
US-13	As a user, I want to clear all my saved data with one action, so that I can reset the app to its default state.	Low	Done

Note: In a real project, these user stories would be entered as cards/tickets in Jira or Trello, assigned to sprints, and tracked through the development lifecycle.